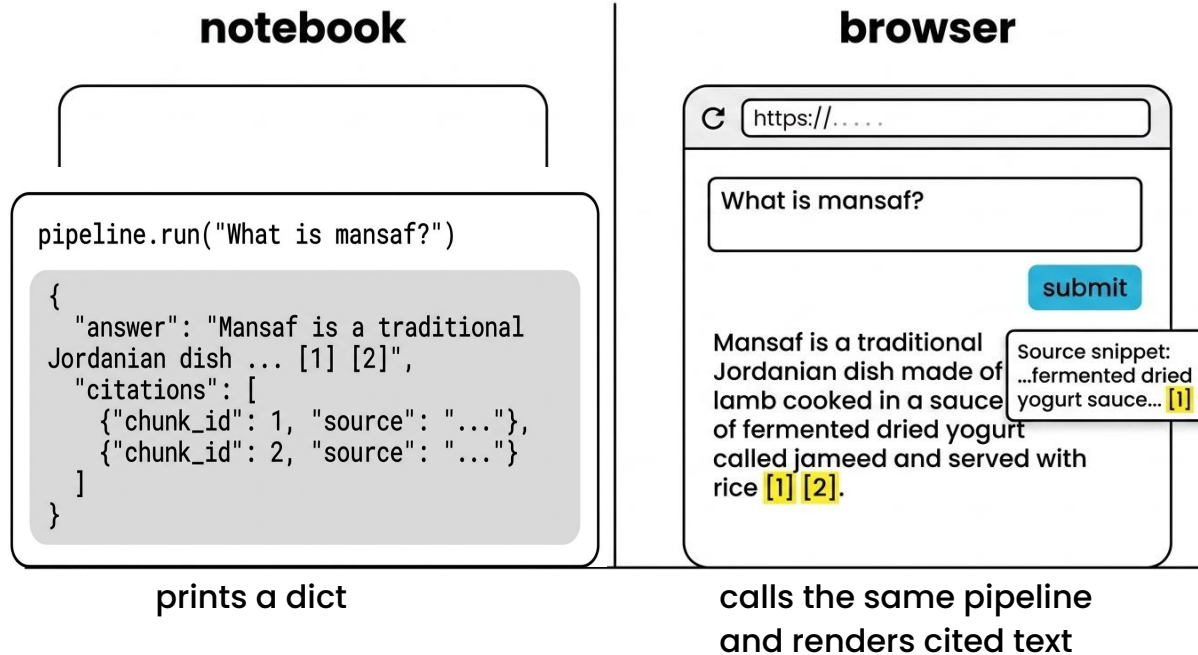


From Notebook To Stack

Module 10 — Typed APIs, the RAG Wrap, Containers, Layout

Open floor for questions instead of
a set “fun fact”

Same Pipeline in Two Places



Four New Things



network



request/
response
contract

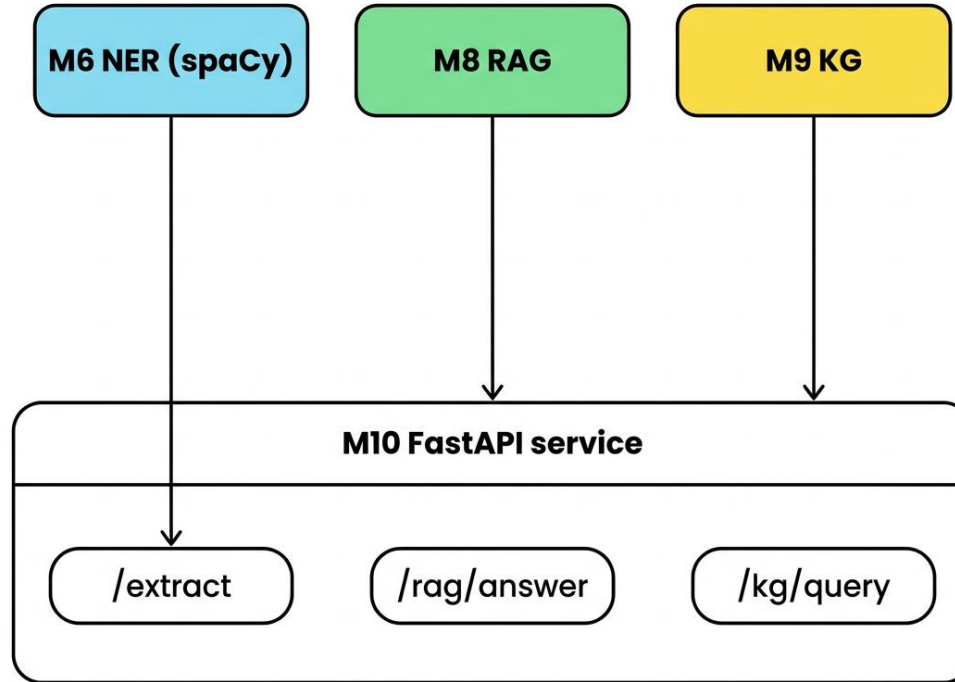


lifecycle

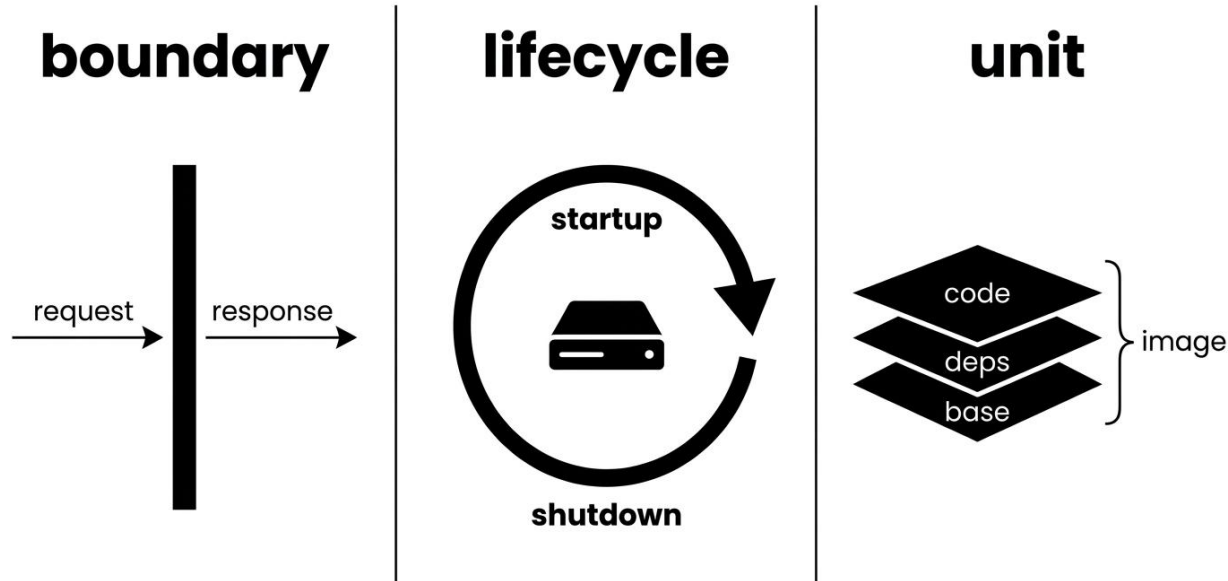


browser

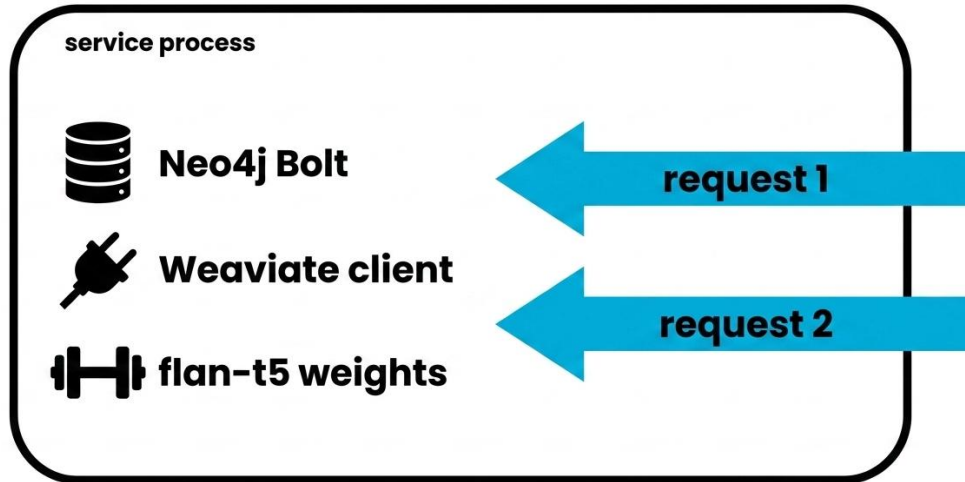
What You Already Know



Deployment

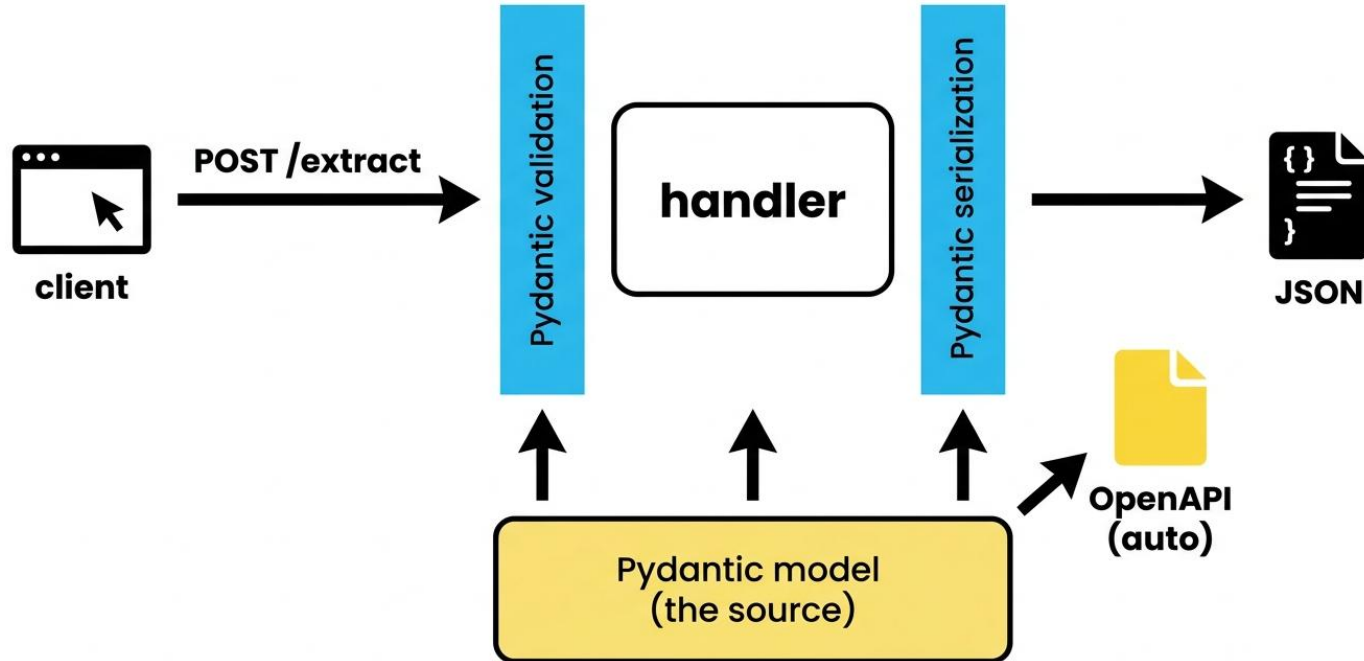


The Lifecycle



resources persist; requests are transient

FastAPI Service: Backend Framework



The Endpoint Checks the Request

validation inside the function

```
if not text:
if len > 5000:
if lang not in {...}:

... processing logic ...
... more code ...
```

inside the function — manual

Pydantic class fast requests

```
if not isinstance(text, str): raise TypeError
if not text: raise ValueError
if len(text) > 1000: raise ValueError
if not language: raise ValueError
if language not in ["en", "es", "fr"]: raise ValueError
```

validation at the boundary

Pydantic

```
if not text:
if len > 5000:
if lang not in {...}:

... clean processing ...
```

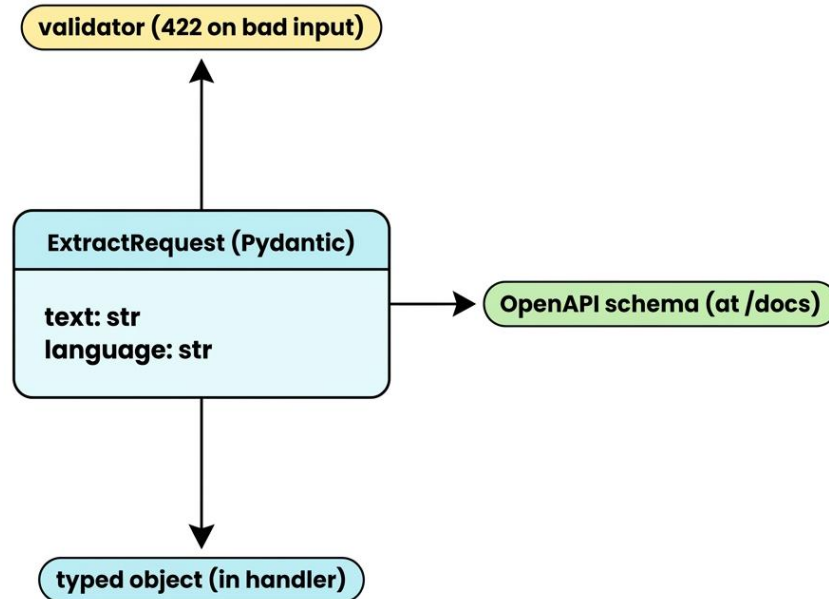
at the boundary — declarative

Pydantic schema:: validation

 validator generated automatically

```
class ExtractRequest(BaseModel):
    text: str = Field(min_length=1, max_length=1000)
    language: str = "en"
```

One Pydantic Model



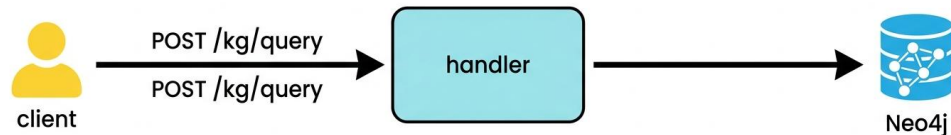
Same model —
validator at the door,
serializer at the exit,
schema in the docs.

Three Wrap Shapes

stateless (NLP)



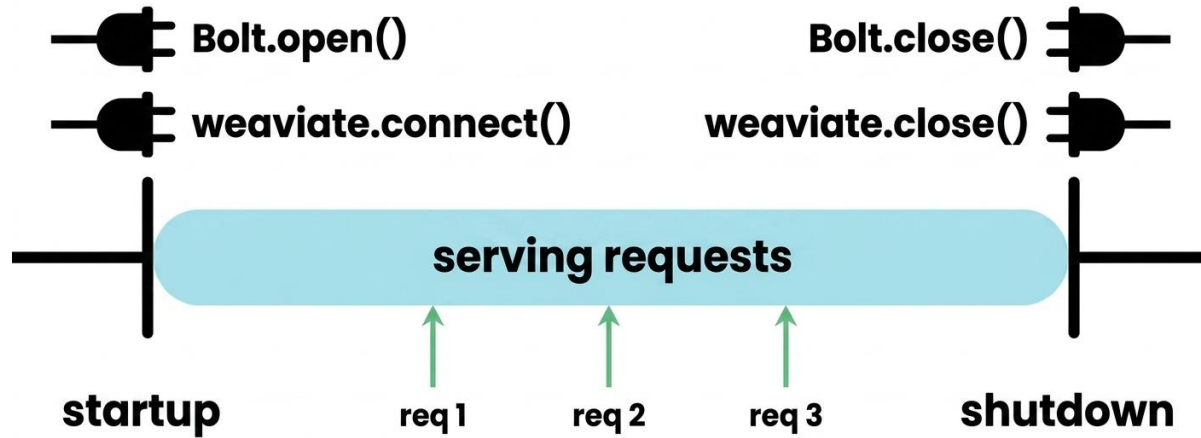
stateful single (KG)



stateful two + model (RAG)



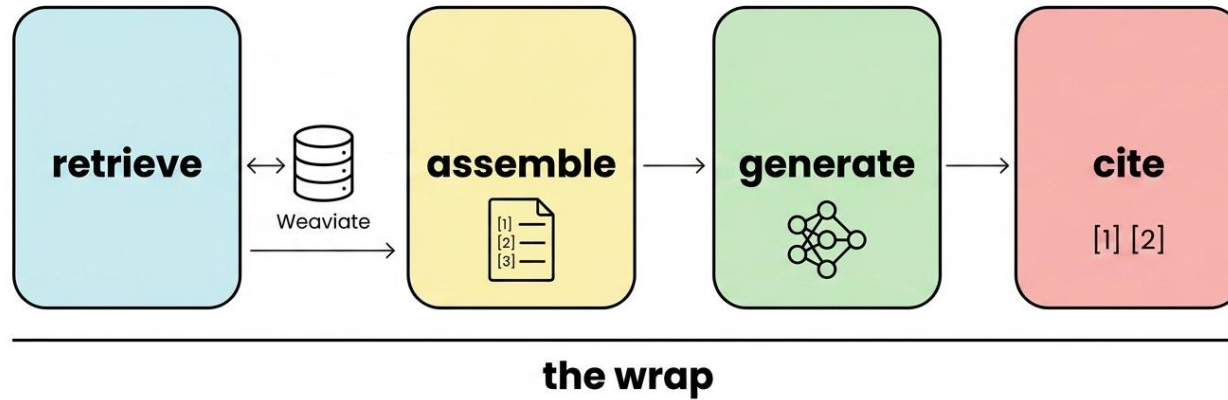
Lifespan



Flask and FastAPI Comparison

Flask		FastAPI	
<code>@app.route('/x', methods=['POST'])</code>	→	<code>@app.post('/x')</code>	→
<code>request.get_json()</code>	→	<code>request: ExtractRequest</code>	→
manual swagger.yaml	→	<code>/docs (auto)</code>	→

RAG As A Wrap

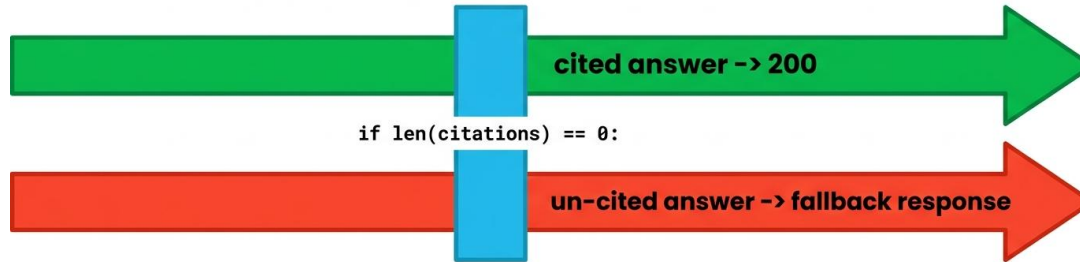


The Endpoint Checks the Citations

inside the model – hope



at the boundary – enforce



The generator may fail to cite, but the endpoint boundary refuses if it does.

A Cited Answer

question

What spices go into mansaf?

**prompt
assembled**

[1] chunk text...
[2] chunk text...
[3] chunk text...

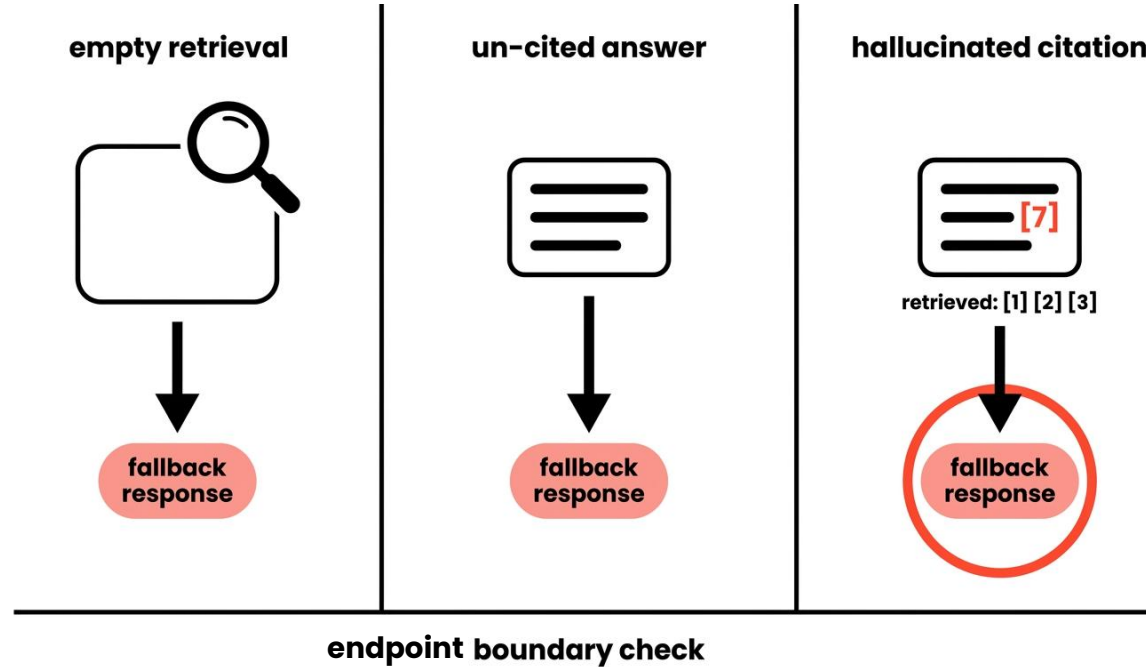
generated

Mansaf uses cardamom, cumin, and turmeric [1].
Some variants add cinnamon and bay leaf [3].

**citations
resolved**

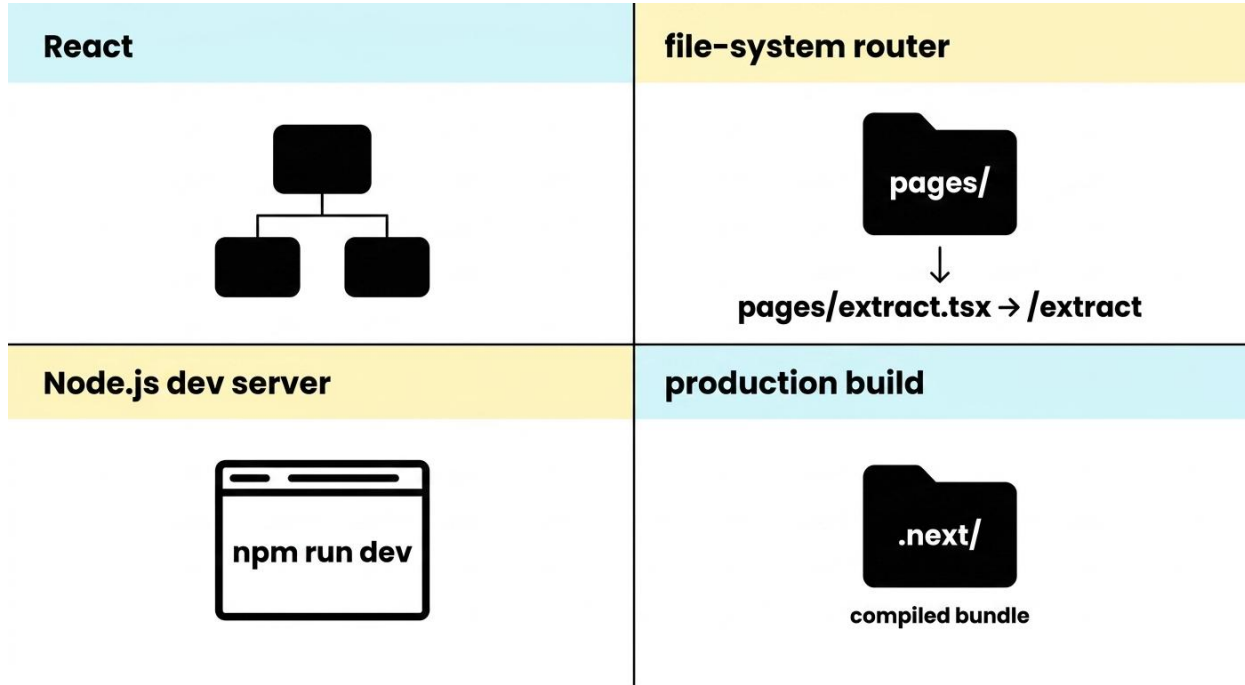
[1] → chunk_id_a
[3] → chunk_id_c

Three Failure Modes



What the Frontend (Browser) Needs from the Backend (FastAPI)

Next.js Frontend



The Mirror

FastAPI route

```
@app.post('/extract')  
def extract(req: ExtractRequest):
```

Pydantic model

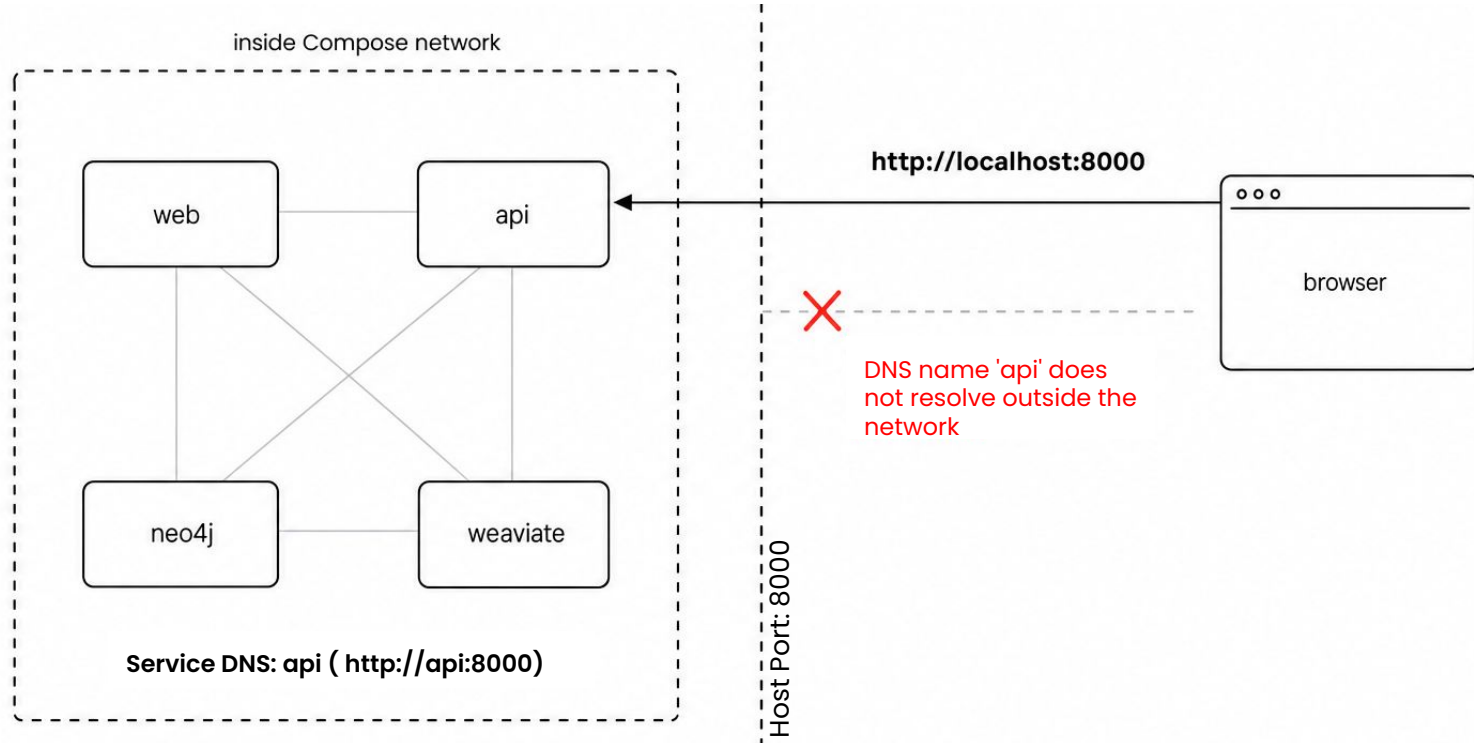
```
class ExtractRequest(BaseModel):  
    text: str  
    language: str
```

TypeScript interface

```
interface ExtractRequest {  
    text: string;  
    language: string;  
}
```

mirror discipline →

Container versus Browser Host



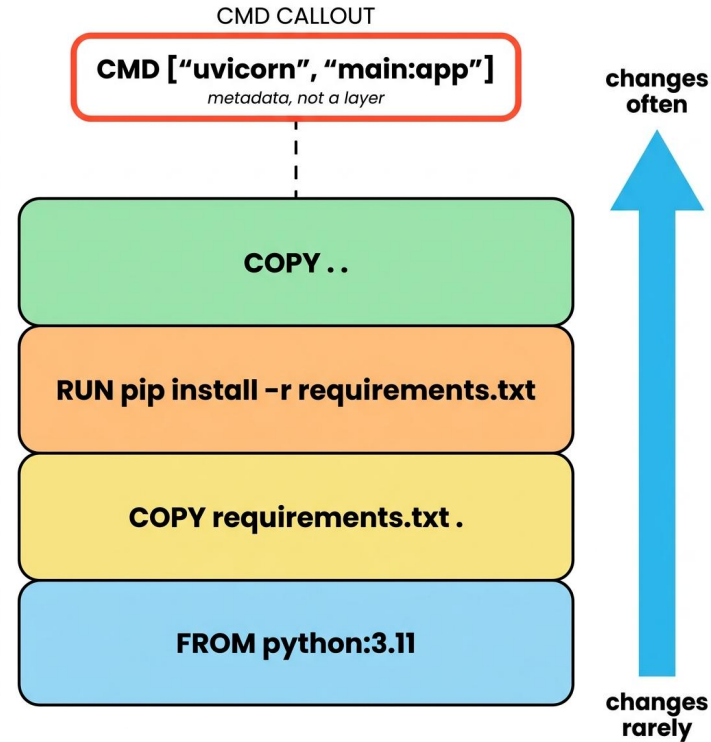
The Browser's Three Pages

/extract	/kg	/rag
 submit PERSON ORG LOC	 submit <pre>graph LR; Rania((Rania)) -- worksat --> TechCorp((TechCorp)); TechCorp -- livesin --> Amman((Amman))</pre>	 submit Based on the data, Jordan [1] has developed a robust tech ecosystem with significant contributions [2].

each page is one backend endpoint

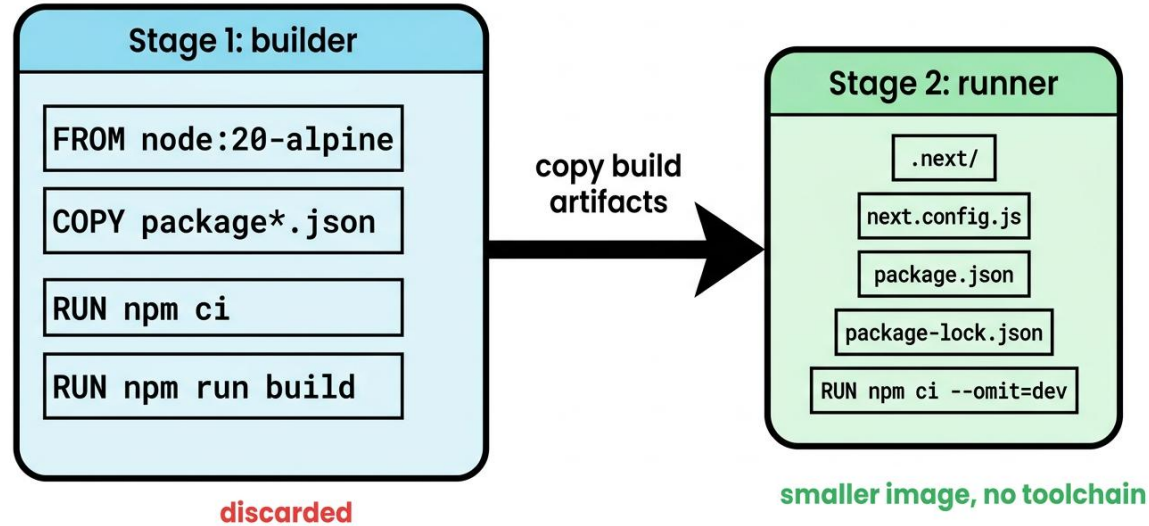
Docker Image, Not File

**Why order matters: Docker reuses
cached layers below the first change.**

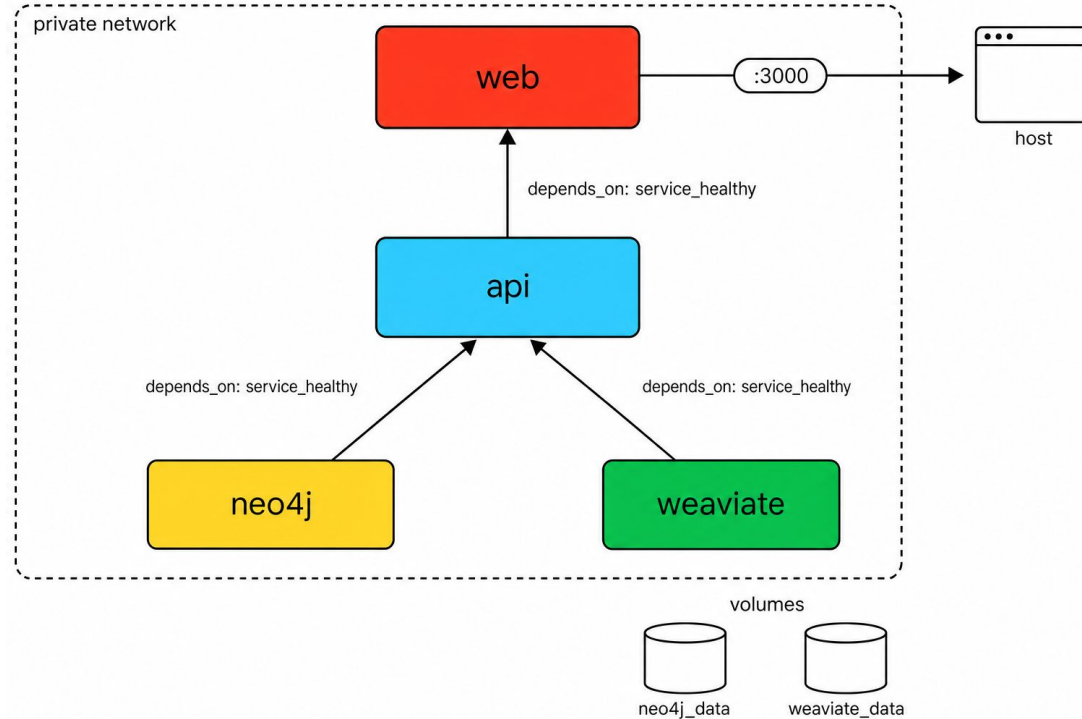


Multi-Stage For Node

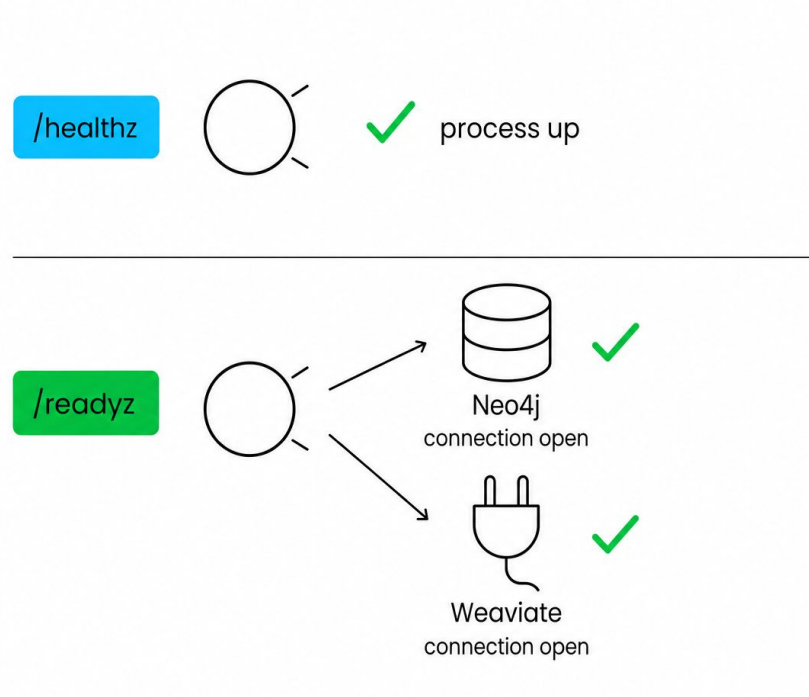
Horizon two stage Node Dockerfile



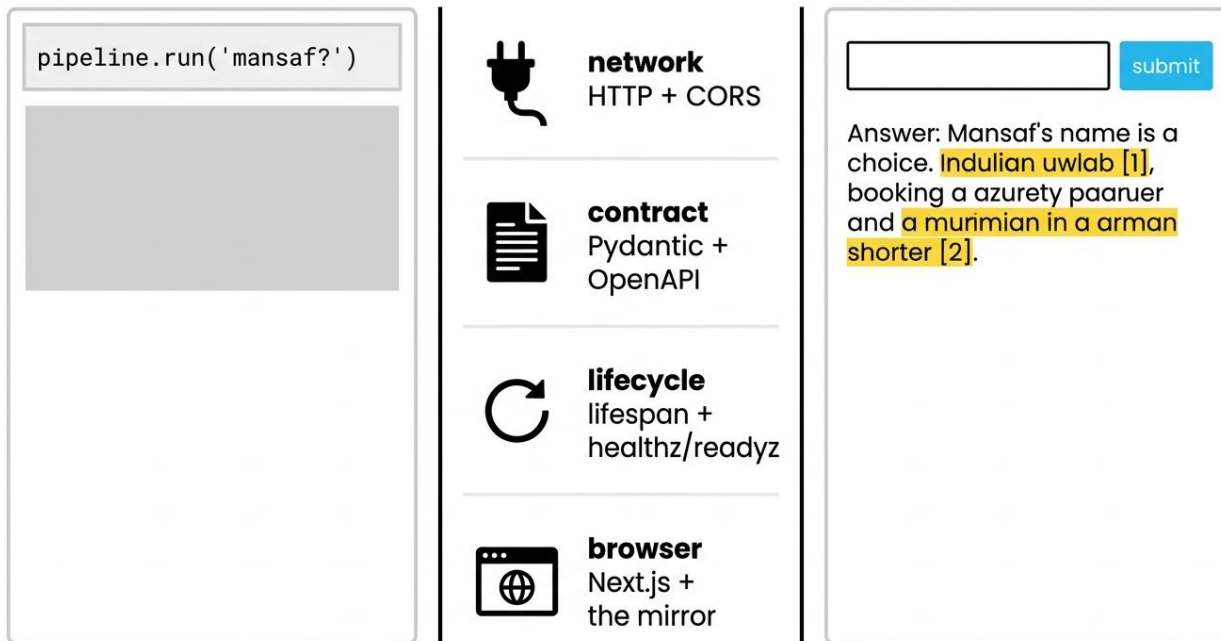
Compose Is The Layout



Readiness Versus Liveness



Summary: Notebook to the Browser



Questions